Toolkit: Chemistry of the Earth

Practical Four: Organic Chemistry and the Carbon Cycle

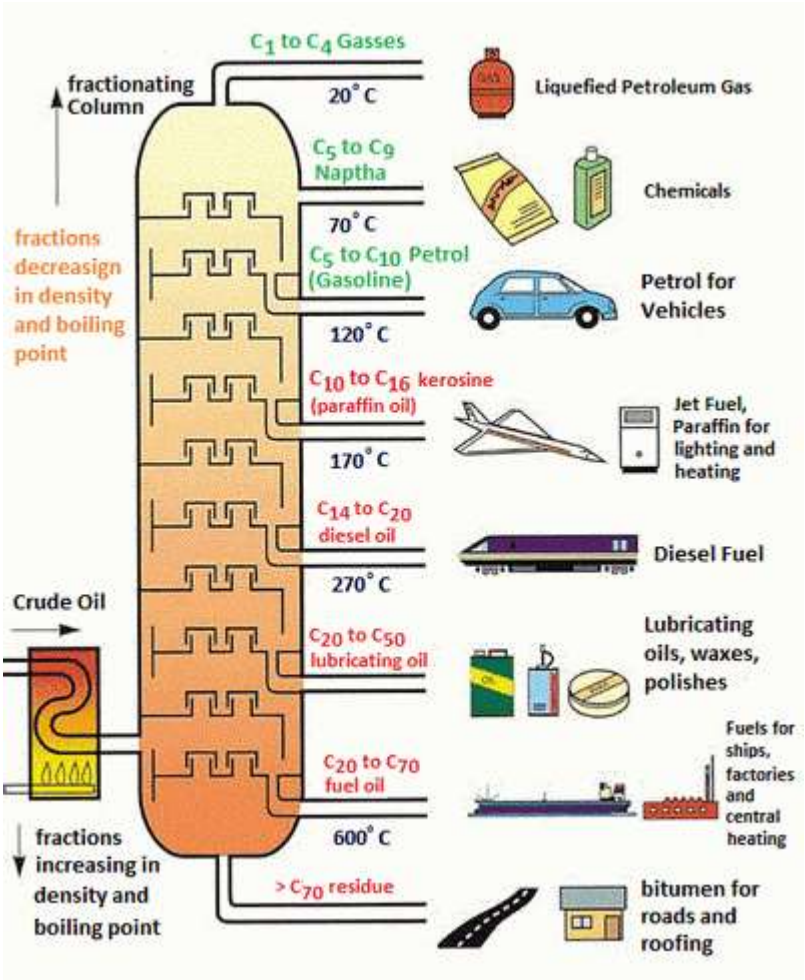
Objectives:

- Familiarization with the notation of organic chemistry
- Investigation of organic reactions in the petroleum industry, this being related to the anthropogenic (human) contribution to the Carbon Cycle

---

FEEDBACK

Oil Refinery:

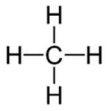
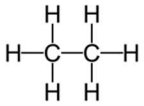
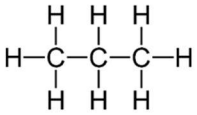
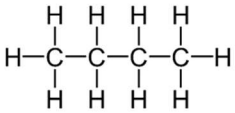
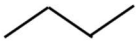


https://www.erl.com.bd/refiningprocess.php

A. Organic molecules and functional groups

1. One of the basic structures in organic chemistry is that of **chain hydrocarbons**. The table below gives information on the first four simple hydrocarbon chain molecules, including formulae and structures.

a) Complete the table for all the molecules listed. In the final column, give the chemical shorthand in which no C or H symbols are used. (This is known as the structural formula.)

Chain structures (Aliphatic compounds)				
No. C atoms	Name	Formula	Structure	Shorthand
1	Methane	CH_4		(none – a dot)
2	Ethane	C_2H_6		—
3	Propane	C_3H_8		
4	Butane	C_4H_{10}		

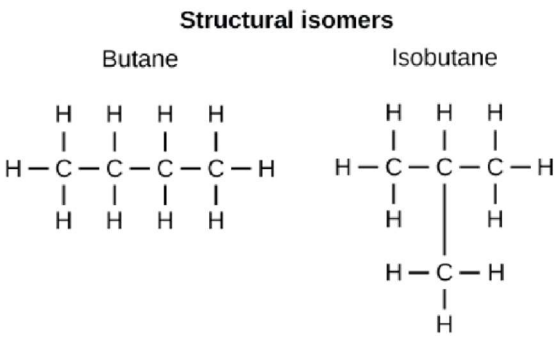
b) Based on the above table, what is the general formula to describe a chain hydrocarbon? ($C_nH_?$)

C_nH_{2n+2}

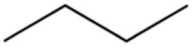
- and hence, what is the formula for octane, C8?

C_8H_{18}

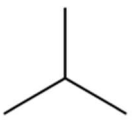
Two compounds with the same chemical formula but a different arrangement of atoms are known as structural **isomers**. For example, butane has two isomers:



c) Draw the shorthand for these isomers:

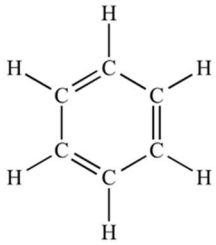
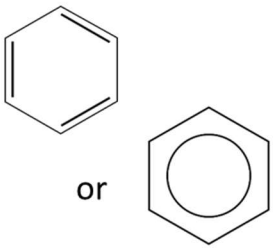
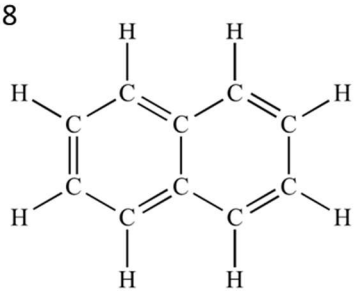
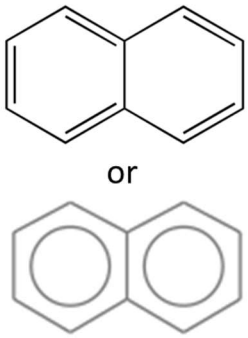


Butane



Isobutane

2. The other basic structure in organic chemistry consists of **rings**, based on the benzene ring which is a single ring. a) The structure of benzene is shown below. What is its chemical formula?

Ring structures (Aromatic compounds)				
No. of rings	Name	Formula	Structure	Shorthand
1	Benzene	C_6H_6		 or
2	Napthalene	$C_{10}H_8$		 or

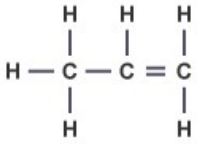
b) A double ring is known as naphthalene. Again, what is its chemical formula?

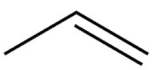
c) For both molecules, draw the chemical shorthand in the final column of the table. Draw two versions for each – one showing the bonding, and one showing the rings of delocalized electrons.

d) Remember that single lines represent single bonds and double lines represent double bonds. How many electrons are shared in each case?

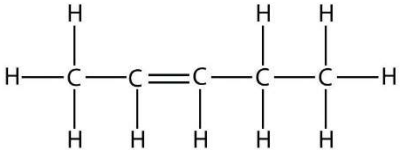
Single bonds: 2 shared electrons Double bonds: 4 shared electrons

3. Chain hydrocarbons can have **double bonds** within the chain. This changes the name to -ENE rather than -ANE. An example is propene:

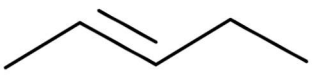




What is the structure of 2-pentene? (in this case the double bond begins on the 2nd C atom)



2-Pentene



4. Functional groups. Simple organic molecules are often modified by functional groups which make them reactive. Some common functional groups are shown in the table below. In an organic molecule, the functional group is attached to a carbon atom, usually by a single bond. The name of the molecule is based on the name of the chain or ring structure, plus the functional groups.

Functional Groups: White, Fig. 12.6

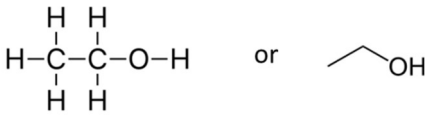
Group	Name	Resulting Compound
—OH	*Hydroxyl*	*Phenols (Aromatics)* *Alcohols (Aliphatics)*
—O—	*Oxo*	*Esters*
R—C(=O)—	*Carbonyl*	*Aldehydes* *Ketones, Quinones*
—C(=O)—OH	*Carboxyl*	*Carboxylic Acid*
—NH_2	*Amino*	*Amines*
—C(=O)—NH_2	*Amido*	*Amides*
—SH	*Thio*	*Thiols*

a) Notice that O and S only form two bonds, and N forms three bonds. How does this fit with the positions of these elements in the periodic table? (think about electron configurations)

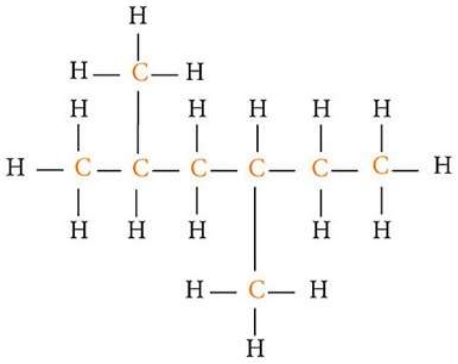
N has 3 p electrons ($1s^2 2s^2 2p^3$) so it needs another 3 electrons to fill its p orbitals.

O and S both have 4 p electrons ($1s^2 2s^2 2p^4$ for oxygen, $1s^2 2s^2 2p^6\ 3s^2 3p^4$ for S), so they both need another 2 electrons to fill their p orbitals.

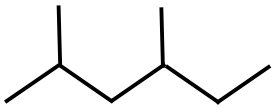
Examples:



This molecule is ethyl alcohol, also known as ethanol. “Ethan” means it is based on a chain of 2 carbon atoms. Alcohol means that there is an –OH functional group.



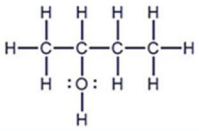
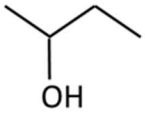
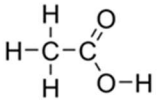
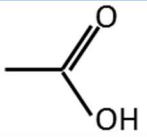
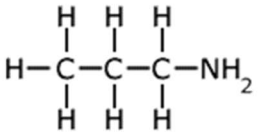
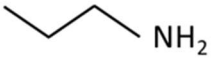
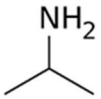
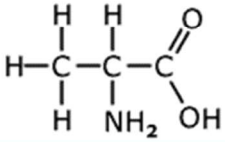
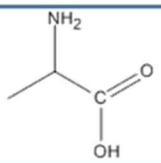
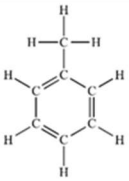
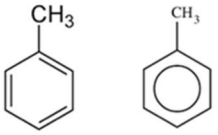
2,4-dimethylhexane



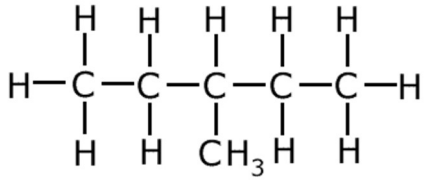
The molecule shown here is 2,4-dimethylhexane.
Hexane means the molecule is based on a chain of 6 carbon atoms.
Dimethyl means there are two methyl groups attached to that chain.
2,4 tells us that those two methyl groups are attached to the 2nd and 4th carbon atoms of the chain.

b) Write the shorthand for 2,4-dimethylhexane in the space to the right of the molecule.

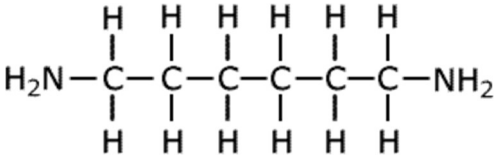
c) Complete the table below to show the structure of the compounds listed: (include shorthand)

Compounds with functional groups		
Name	**Structure**	**Shorthand**
2-Butanol (butane with an alcohol group on its 2nd C atom)		
Acetic acid (methane with a carboxylic acid group)		
Propylamine (propane with an amine functional group)		 Isomer: Isopropylamine 
2-aminopropanoic acid (propane with an amino group on the 2nd C atom and a carboxylic acid group on a C atom at the end of the chain)		
Methyl benzene (also named toluene) (benzene with a methyl group)		

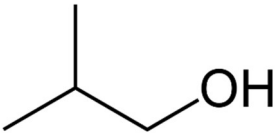
d) What are the names of the following molecules?



3-methyl pentane



1,6-diaminohexane



2-methyl propanol
(this is also isobutanol: see Q. 1c)

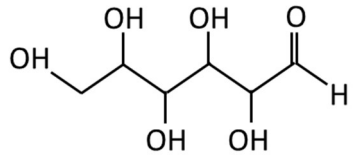
4. Some simple organic molecules with **biological importance.**

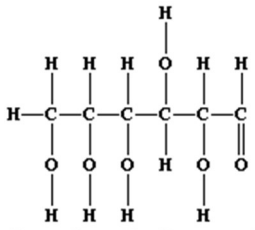
a) Sugars. Sugars are essential sources of energy in organisms. *Glucose* is a simple sugar, made during photosynthesis from water and carbon dioxide, using energy from sunlight. It circulates in the blood of animals as blood sugar, and supplies almost all of the energy for the brain. Glucose is the most important source of energy for cellular respiration:

Respiration: $CH_2O + O_{2\,(g)} = CO_{2\,(g)} + H_2O_{(l)} + \text{energy}$

Simple sugars like glucose are named monosaccharides. They have the general formula $C_nH_{2n}O_n$ with n between 3 and 7. This is why the respiration equation shows sugar as CH_2O.

Glucose (open chain form):



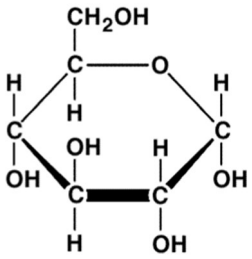


Structural formula of glucose ($C_6H_{12}O_6$)

Write out the complete structure of (open-chain) glucose, determine its chemical formula, and show that it has the general formula given above.

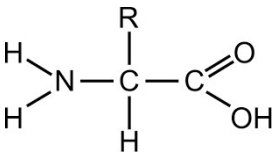
Most glucose exists in a ring form:

Check that this has the same formula.

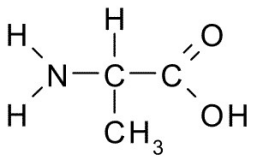


b) Amino acids. Amino acids are essential ingredients of biological systems. They form the building-blocks of a vast array of proteins and participate in processes such as neurotransmitter transport and biosynthesis. There are 20 amino acids in the genetic code (DNA/RNA).

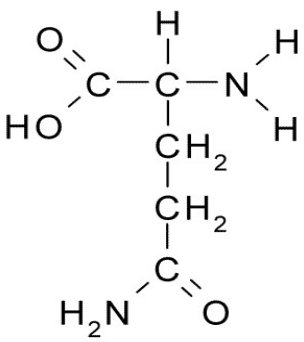
Amino acids are defined by the following general formula, where R represents different units. For example, one of the simplest amino acids is *alanine,* where R is simply a methyl group:



amino acid general formula

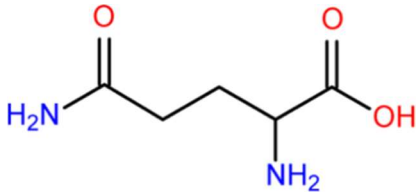


alanine



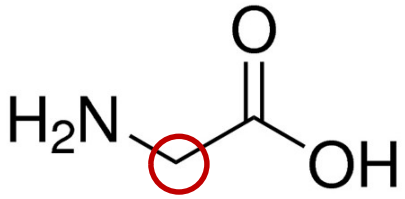
Glutamine (left) is used by almost all living beings in the biosynthesis of proteins, as well as diverse other functions.

Write the shorthand formula for glutamine:

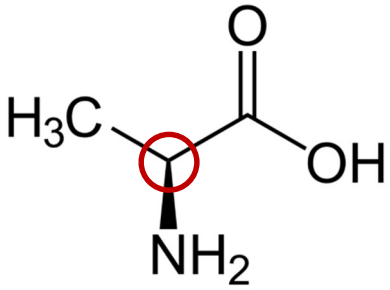


c) Chirality

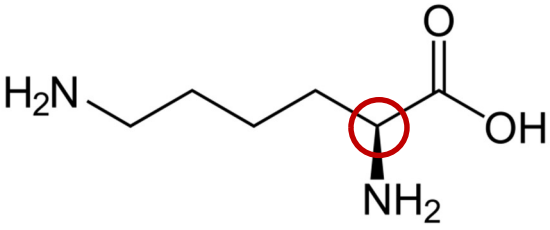
Most amino acids are chiral molecules: there are two forms that are mirror images of each other, and these two molecules cannot be superposed. For the three amino acids below, identify where in the structure you find the carbon atom attached to "R" (see above). Then decide whether each molecule is a chiral molecule.



Glycine



Alanine



Lysine

The carbon atom attached to the "R" group is circled in each case. Glycine is the only non-chiral amino acid. For glycine, there are two H atoms attached to the circled carbon atom, so the mirror images will superpose. For other amino acids, the circled carbon atom has four different groups attached (H, COOH, NH_2 and the R group), so these molecules are chiral.

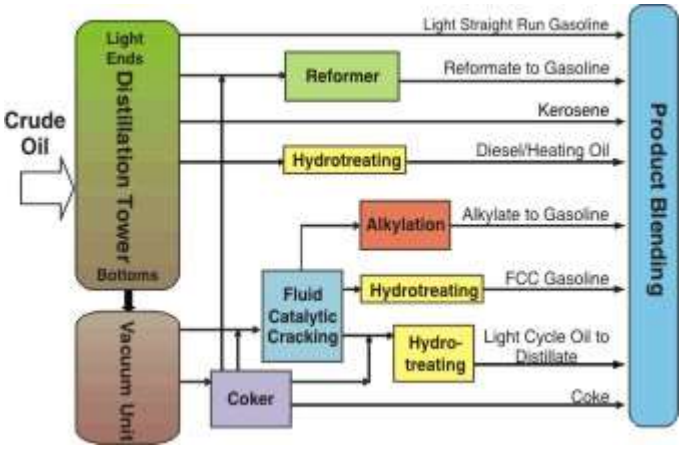
B. Organic Chemistry and Oil Refineries.

1. Distillation. The diagram on the front page shows the basic principles of an oil refinery. Crude oil, consisting of very long-chain hydrocarbons, is broken down through the process of distillation to produce a variety of different products consisting of shorter chain molecules.

a) What is the basis for the distillation process? How and why does the chemistry of the distillation product change as you go upwards through the refinery plant?

First, crude oil, which is a mixture of many different hydrocarbon molecules, is heated so that it all vaporizes. The gas passes into the distillation plant and condenses on plates that are held at different temperatures, as it rises. Longer chain molecules have high density and higher boiling points / higher condensation temperatures – these condense at the bottom of the column. Shorter chain molecules have lower density and lower boiling points / lower condensation temperatures – these rise to the top of the column where they are condensed and collected. Products go from solids (waxes, polishes) to liquids (fuels) to gases (image shows a canister of compressed gas) with height in the column.

In fact, refineries are more complex because chemical processes are carried out on the different distillates to make them more useful. The diagram below shows some of these processes, including cracking, reforming, and alkylation. We will look at some of the reactions associated with these processes, as well as isomerization and polymerization.



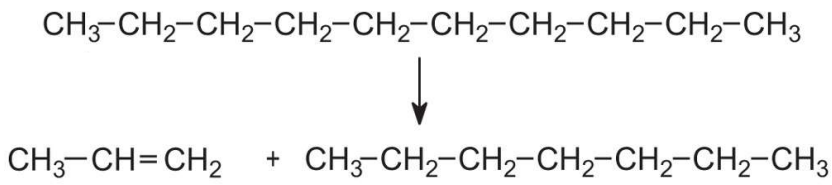
Basic Petroleum Refining Process. (http://www.cpp.org.au/Refinery.php)

Much of the text below is taken from The Essential Chemical Industry - Online: http://www.essentialchemicalindustry.org/processes/cracking-isomerisation-and-reforming.html

Use your knowledge of organic chemistry to understand this text fully!

2. Cracking and Anti-knock compounds

Cracking, as the name suggests, is a process in which large hydrocarbon molecules are broken down into smaller and more useful ones, for example:



a) Write this reaction in words, i.e. just write the names of the different compounds:
(Hint: for naming the long molecules, a clue is in the names of athletics competitions consisting of multiple track and field events.)

decane → propene + heptane

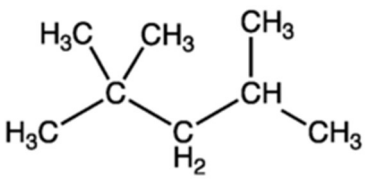
Note that the name propene refers to the presence of a double bond in the molecule that is based on propane. Hept- is a prefix for 7, and dec- is a prefix for ten, as in heptathlon and decathlon.

The mixture of C5-C10 hydrocarbons obtained directly from the distillation of crude oil contains a high proportion of straight-chain alkanes. However, if this mixture is used as petrol, it does serious damage to a car's engine. Petrol containing a high proportion of straight chain alkanes tends to ignite in the cylinder of the car engine as the piston increases the pressure and before the cylinder reaches the optimum position. Ideally, the mixture of petrol vapour and air is ignited with a spark at a predetermined position of the piston in the cylinder. This problem of premature ignition is referred to as pre-ignition and also as engine knock. The term knock is used, as pre-ignition can be heard. Severe knock can cause serious engine damage. However, branched-chain alkanes, cycloalkanes and aromatic hydrocarbons are much more resistant to knock. Straight-chain alkanes are converted into these compounds in a series of processes in the refinery.

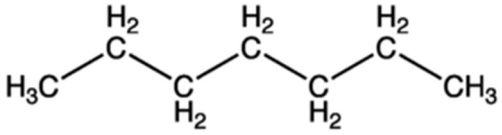
The resistance of petrol to knock is measured in terms of an octane rating (octane number). The higher the number, the less likely is a fuel to pre-ignite. The octane rating is on a scale where heptane is given an arbitrary score of 0 and 2,2,4-trimethylpentane (iso-octane) one of 100. Thus, a petrol with the same knocking characteristics as a mixture of 95% 2,2,4-trimethylpentane and 5% heptane has an octane rating of 95. A rating of 95 does not mean that the petrol contains just iso-octane and heptane in these proportions, but that it has the same tendency to knock as this mixture. The octane ratings of petrols usually available for cars range from 95 upwards and contain a mixture of straight-chain, branched, cyclic and aromatic hydrocarbons, produced by cracking and other processes described below.

b) Draw the structure of the molecules mentioned above:

2,2,4-trimethylpentane (also named iso-octane)
(Pentane has 5 C atoms)

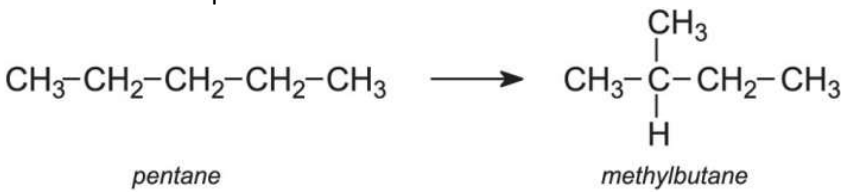


Heptane (7 C atoms)

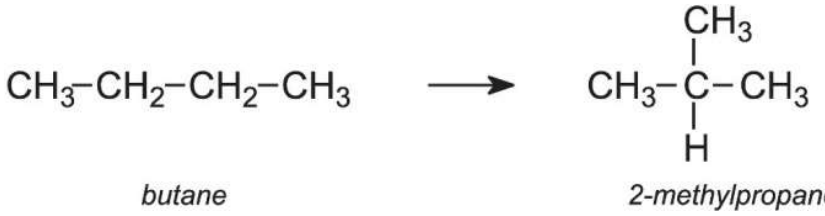


3.Isomerization

Isomerisation is the process in which hydrocarbon molecules are rearranged into a more useful isomer. For example:



The process is particularly useful in enhancing the octane rating of petrol, as branched alkanes burn more efficiently in a car engine than straight-chain alkanes. An important example is as follows:



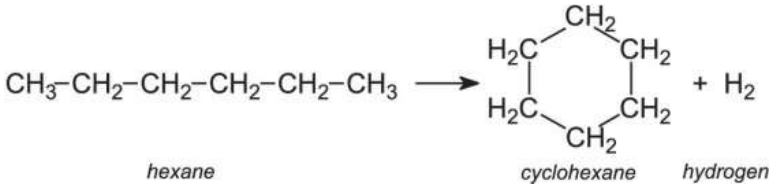
(See Q. 1b, page 2)

a) Name the compounds in the above reactions.

The molecule on the left is also known as LPG: Liquefied petroleum gas, which is liquefied propane and butane. See figure on Page 1. These molecules are used as fuels.

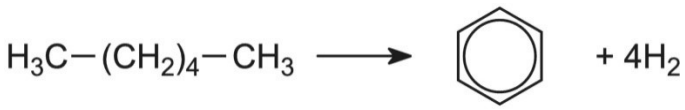
4. Reforming

Reforming is another process in which hydrocarbon molecules are rearranged into other molecules, usually with the loss of a small molecule such as hydrogen. An example is the conversion of an alkane molecule into a cycloalkane or an aromatic hydrocarbon:



This is a very important process for the petroleum and chemical industries. It enables straight chain alkanes to be converted into branched-chain alkanes, cyclohexanes and aromatic hydrocarbons which are used to enhance the octane number of petrol.

Another example is the reaction in which hexane is reformed to benzene. Again, hydrogen is a reaction product. Write the reaction that takes place: draw the formulae of the organic molecules and balance the reaction to determine how many moles of hydrogen are produced for each mole of hexane that reacts:

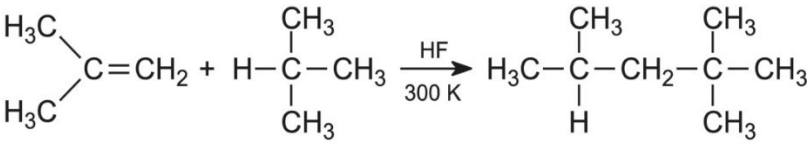


(4 moles of hydrogen for each mole of hexane)

5. Alkylation

Alkylation is the transfer of an alkyl group from one molecule to another. In a refinery, alkylation refers to the alkylation of alkanes, for example, 2-methylpropane (isobutane) with alkenes, in the presence of a strong acid catalyst* such as hydrofluoric acid or sulfuric acid. The reaction is carried out at mild temperatures (between 273 and 303K). Cooling is needed as the reaction is exothermic.

The product from 2-methylpropane and 2-methylpropene (isobutene) is a mixture of branched-chain alkanes, mainly 2,2,4-trimethylpentane (isooctane):

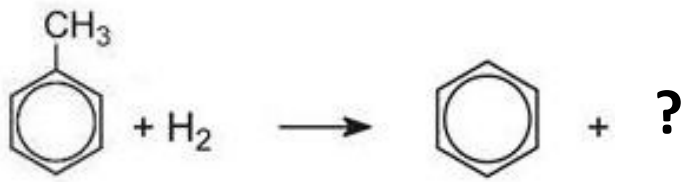


2,2,4-trimethylpentane

Why is this reaction important? (See Qs 2, 3, and 4 above!)

The reaction product is an important anti-knocking agent.

Dealkylation: The opposite of alkylation occurs when methylbenzene (toluene) is heated with hydrogen over a catalyst. The dealkylation of methylbenzene produces benzene, a more valuable product in the chemical industry. In this reaction (below), what is the other reaction product? – give it a formula and a name. Is it a solid, liquid, or gas?



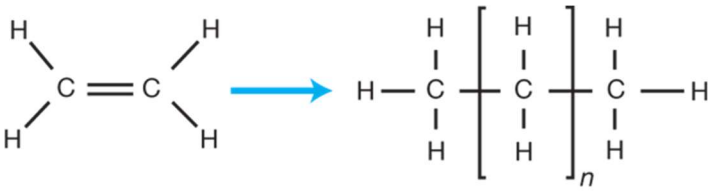
Reaction product is CH_4, methane – a gas.

*A catalyst is a substance that causes or accelerates a chemical reaction without itself undergoing any permanent chemical change.

6. Polymerization

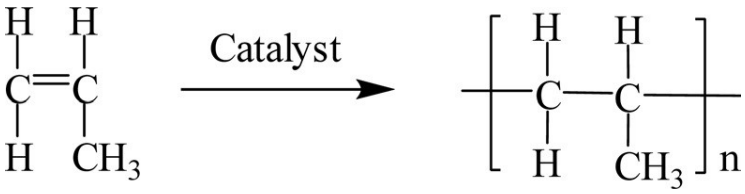
When many molecules of a simple compound join together, the product is called a polymer and the process is termed polymerization. The simple compounds whose molecules join together to form the polymers are called monomers. Monomers are generally produced in the refinery and the polymers, such as poly(ethene) and poly(propene), are often manufactured nearby. The major differences between smaller molecules and polymers lie not with their chemical properties but with their physical ones. Their larger sizes lead to much stronger intermolecular forces, leading in turn to much higher melting points, and the characteristic properties of hardness and flexibility found in plastics.

Examples:



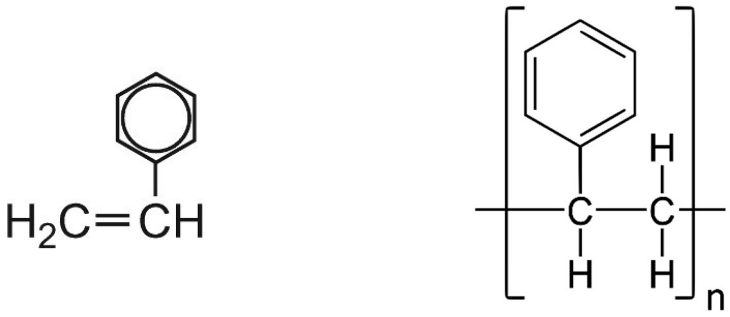
Ethylene Polyethylene

n is a very large number – thousands of monomer units combine in a polymer



Propylene *Polypropylene*

Given the formula for phenylethene below, predict the formula for polyphenylethene, otherwise known as polystyrene:



Phenylethene *Polyphenylethene, or polystyrene*

7. Burning Fuel: Production of CO_2

In a fuel-burning reaction, a hydrocarbon reacts with oxygen, producing carbon dioxide and water. As an example, complete the reaction below by filling in the chemical formula for each compound. Balance the reaction if necessary.

$$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$$

Methane + Oxygen → Carbon dioxide + Water

Now write and balance comparable reactions for the following fuels:

Octane (gasoline)

$$2\ C_8H_{18} + 25\ O_2 = 16\ CO_2 + 18\ H_2O$$

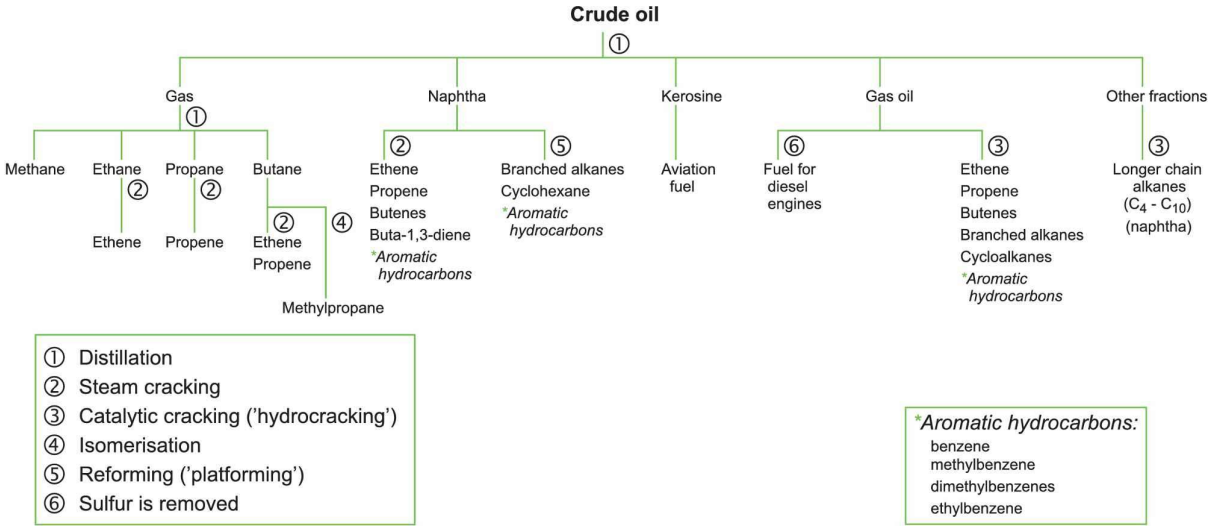
Ethanol

$$C_2H_5OH + 3\ O_2 = 2\ CO_2 + 3\ H_2O$$

These reactions show how CO_2 is the product of burning fossil fuels. CO_2 is a gas, so it enters the atmosphere, where it contributes to the greenhouse effect.

SUMMARY

The figure below summarizes the crude oil distillation and refining process. Read over it and reflect on the terminology we have covered in this practical, which provides an overview of the organic chemistry of fossil fuel use.



http://www.essentialchemicalindustry.org/processes/cracking-isomerisation-and-reforming.html